3.1 Measure and integration: solutions

Read a solution only after a serious attempt. After reading, close the file and write the proof out yourself.

Warm-up

1.

(a) Let $\{\mathcal{M}_i\}_{i\in I}$ be σ-algebras on X and $\mathcal{M} = \bigcap_i \mathcal{M}_i$. Each $\mathcal{M}_i$ contains $\emptyset$, so $\emptyset \in \mathcal{M}$. If $A \in \mathcal{M}$ then $A \in \mathcal{M}_i$ for all i, so $A^c \in \mathcal{M}_i$ for all i, so $A^c \in \mathcal{M}$. If $A_1, A_2, \cdots \in \mathcal{M}$ then for each i all $A_n \in \mathcal{M}_i$, so $\bigcup_n A_n \in \mathcal{M}_i$; hence $\bigcup_n A_n \in \mathcal{M}$.

(b) Let $\mathcal{M}_1 = \{\emptyset, \{1\}, \{2,3\}, X\}$ and $\mathcal{M}_2 = \{\emptyset, \{2\}, \{1,3\}, X\}$. Both are σ-algebras. Their union contains $\{1\}$ and $\{2\}$ but not $\{1,2\}$, so it is not closed under unions.

2.

Put $F_n = E_1 \setminus E_n$. Then $F_1 \subseteq F_2 \subseteq \cdots$ and $\bigcup_n F_n = E_1 \setminus \bigcap_n E_n$. By continuity from below, $\mu(E_1 \setminus \bigcap E_n) = \lim \mu(E_1 \setminus E_n)$. Since $\mu(E_1) < \infty$, all sets involved have finite measure and $\mu(E_1 \setminus A) = \mu(E_1) - \mu(A)$ for $A \subseteq E_1$. So $\mu(E_1) - \mu(\bigcap E_n) = \mu(E_1) - \lim \mu(E_n)$; cancel the finite quantity $\mu(E_1)$.

Counterexample: $E_n = [n, \infty) \subseteq \mathbb{R}$ with Lebesgue measure. Each $m(E_n) = \infty$, but $\bigcap_n E_n = \emptyset$ has measure 0.

(Continuity from below, for completeness: if $A_1 \subseteq A_2 \subseteq \cdots$, let $B_1 = A_1$, $B_k = A_k \setminus A_{k-1}$; these are disjoint with $\bigcup_{k\le n} B_k = A_n$, so $\mu(\bigcup A_n) = \sum_k \mu(B_k) = \lim_n \mu(A_n)$.)

3.

The limit is 1.

Let $f_n(x) = (1-x/n)^n \chi_{[0,n]}(x)$ on $[0,\infty)$. For fixed $x \ge 0$ and $n > x$, $f_n(x) = (1-x/n)^n \to e^{-x}$. Since $1 - t \le e^{-t}$ for all real t, for $0 \le x \le n$ we have $0 \le 1 - x/n \le e^{-x/n}$, and raising to the n-th power gives $0 \le f_n(x) \le e^{-x}$. The function e^{-x} is integrable on $[0,\infty)$, so by the dominated convergence theorem

$$\int_0^n \left(1 - \frac{x}{n}\right)^n dx = \int_0^\infty f_n \to \int_0^\infty e^{-x}\, dx = 1.$$

Check: directly, $\int_0^n (1 - x/n)^n\, dx = \frac{n}{n+1} \to 1$.

4.

(a) Let $A = \{a_1, a_2, \dots\}$ and $\varepsilon > 0$. The intervals $I_k = (a_k - \varepsilon 2^{-k-1}, a_k + \varepsilon 2^{-k-1})$ cover A with total length $\sum_k \varepsilon 2^{-k} = \varepsilon$. So $m^*(A) \le \varepsilon$ for every ε, hence $m^*(A) = 0$, and null sets are measurable.

(b) $C = \bigcap_k C_k$ where C_k is a union of 2^k disjoint closed intervals each of length 3^{-k}. So $m(C) \le m(C_k) = (2/3)^k \to 0$.

Uncountability: every $x \in [0,1]$ has a ternary expansion $x = \sum_{k\geq 1} d_k 3^{-k}$, $d_k \in \{0,1,2\}$. C is exactly the set of x having an expansion with all $d_k \in \{0,2\}$ (at stage k one removes the points whose k-th digit must be 1). The map $(\epsilon_k) \mapsto \sum_k 2\epsilon_k 3^{-k}$ from $\{0,1\}^{\mathbb{N}}$ to C is injective (two different $0/2$-expansions give different numbers: if they first differ at place j, the difference is at least $2\cdot 3^{-j} - \sum_{k>j} 2\cdot 3^{-k} = 3^{-j} > 0$) and onto C. Since $\{0,1\}^{\mathbb{N}}$ is uncountable (Cantor's diagonal argument), so is C.

5.

For $n \geq 1$ let $E_n = \{f \geq 1/n\}$. Then $\frac{1}{n}\chi_{E_n} \leq f$, so $\frac{1}{n}\mu(E_n) \leq \int f = 0$, so $\mu(E_n) = 0$. Since $\{f > 0\} = \bigcup_n E_n$, countable subadditivity gives $\mu(\{f>0\}) = 0$.

6.

Let $E = \{|f| \geq \alpha\}$. Then $\alpha\chi_E \leq |f|$ pointwise, so $\alpha\mu(E) = \int \alpha\chi_E \leq \int |f|$.

If $f \in L^1$, then $\{|f| = \infty\} \subseteq \{|f| \geq n\}$ for every n, so its measure is at most $\frac{1}{n}\|f\|_1 \to 0$.

Core

7.

Define $x \sim y$ on $[0,1]$ iff $x - y \in \mathbb{Q}$. This is an equivalence relation. **Using the axiom of choice**, choose a set $V \subseteq [0,1]$ containing exactly one element of each equivalence class.

Let $q_1, q_2, \ldots$ enumerate $\mathbb{Q} \cap [-1,1]$ and $V_k = V + q_k$.

- *Disjoint:* if $x \in V_j \cap V_k$ then $x - q_j, x - q_k \in V$ are equivalent (their difference $q_k - q_j$ is rational), so they are equal, so $q_j = q_k$ and $j = k$.
- *Cover* $[0,1]$*:* given $x \in [0,1]$, let $v \in V$ be the representative of its class. Then $x - v \in \mathbb{Q} \cap [-1,1]$, so $x - v = q_k$ for some k, and $x \in V_k$.
- *Contained in* $[-1,2]$*:* clear.

Suppose V were measurable. By translation invariance each V_k is measurable with $m(V_k) = m(V)$. By countable additivity and monotonicity,

$$1 = m([0,1]) \leq m\left(\bigcup_k V_k\right) = \sum_{k=1}^{\infty} m(V) \leq m([-1,2]) = 3.$$

If $m(V) = 0$ the sum is $0 < 1$; if $m(V) > 0$ the sum is $\infty > 3$. Either way a contradiction, so V is not measurable.

8.

(a) By definition of outer measure there are open intervals I_j covering E with $\sum_j |I_j| < m(E) + \varepsilon$. Let $U = \bigcup_j I_j$, an open set containing E. By countable subadditivity $m(U) \leq \sum |I_j| < m(E) + \varepsilon$. Since E is measurable and $m(E) < \infty$, $m(U \setminus E) = m(U) - m(E) < \varepsilon$.

(b) *Bounded case.* Suppose $E \subseteq J = [-N, N]$. Apply (a) to the measurable set $J \setminus E$ (finite measure): there is an open $U \supseteq J \setminus E$ with $m(U \setminus (J \setminus E)) < \varepsilon$. Let $K = J \setminus U = J \cap U^c$, closed and bounded, hence compact. Since $J \setminus E \subseteq U$, we have $K \subseteq E$. Also $E \setminus K = E \cap U$, and every point of $E \cap U$ lies in U but not in $J \setminus E$, so $E \setminus K \subseteq U \setminus (J \setminus E)$ and $m(E \setminus K) < \varepsilon$.

General case. Let $E_N = E \cap [-N, N]$. The sets $E \setminus E_N$ decrease to $\emptyset$ and $m(E) < \infty$, so by continuity from above there is N with $m(E \setminus E_N) < \varepsilon/2$. By the bounded case pick compact $K \subseteq E_N$ with $m(E_N \setminus K) < \varepsilon/2$. Then $m(E \setminus K) < \varepsilon$.

9.

Discard the null set where $f_n \not\to f$ (add it to E at the end). For $k, n \geq 1$ let

$$E_n(k) = \bigcup_{j \geq n} \left\{ x : |f_j(x) - f(x)| \geq \tfrac{1}{k} \right\}.$$

For fixed k, $E_1(k) \supseteq E_2(k) \supseteq \cdots$, and $\bigcap_n E_n(k)$ is the set where $|f_j - f| \geq 1/k$ for infinitely many j, which is empty because $f_j \to f$ everywhere (after discarding). Since $\mu(X) < \infty$, continuity from above gives $\mu(E_n(k)) \to 0$, so choose n_k with $\mu(E_{n_k}(k)) < \varepsilon 2^{-k}$. Let $E = \bigcup_k E_{n_k}(k)$, so $\mu(E) < \varepsilon$. If $x \notin E$, then for every k and every $j \geq n_k$, $|f_j(x) - f(x)| < 1/k$. That is uniform convergence on $X \setminus E$.

Finite measure is needed. On $\mathbb{R}$ let $f_n = \chi_{[n,n+1]}$, so $f_n \to 0$ everywhere. If $m(E) < 1$, then $[n, n+1] \setminus E \neq \emptyset$ for every n, so $\sup_{\mathbb{R} \setminus E} |f_n| = 1$ for every n; there is no uniform convergence off E.

10.

(a) $f_n = \chi_{[n,n+1]}$ on $\mathbb{R}$: $\liminf f_n = 0$ pointwise, so $\int \liminf f_n = 0 < 1 = \liminf \int f_n$.

(b) Suppose $f_n \to f$ a.e., $|f_n| \leq g \in L^1$. Then $|f| \leq g$ a.e., so $f \in L^1$. The functions $h_n = 2g - |f_n - f|$ are non-negative a.e. and $h_n \to 2g$ a.e. Fatou's lemma gives

$$\int 2g \leq \liminf_n \int (2g - |f_n - f|) = \int 2g - \limsup_n \int |f_n - f|.$$

Since $\int 2g < \infty$ we can subtract it: $\limsup \int |f_n - f| \leq 0$. Hence $\int |f_n - f| \to 0$ and $|\int f_n - \int f| \leq \int |f_n - f| \to 0$.

11.

Let $g(x) = \frac{1}{1+x}$, so $g(1) = \frac{1}{2}$ and $|g'(x)| = \frac{1}{(1+x)^2} \leq 1$ on $[0, 1]$. Integrate by parts (both factors are C^1 on $[0, 1]$):

$$\int_0^1 n x^{n-1} g(x)\, dx = [x^n g(x)]_0^1 - \int_0^1 x^n g'(x)\, dx = \frac{1}{2} - \int_0^1 x^n g'(x)\, dx.$$

And $\left| \int_0^1 x^n g'(x)\, dx \right| \leq \int_0^1 x^n\, dx = \frac{1}{n+1} \to 0$. So the limit is $\frac{1}{2}$.

(The general principle: $n x^{n-1}\, dx$ is a probability measure on $[0, 1]$ concentrating at 1, so $\int n x^{n-1} g \to g(1)$ for any g continuous at 1 and bounded.)

12.

(a) Enumerate the dyadic intervals level by level: for $k = 0, 1, 2, \ldots$ and $j = 0, \ldots, 2^k - 1$, let $I_{k,j} = [j2^{-k}, (j+1)2^{-k}]$, and let $f_1, f_2, \ldots$ be the indicators $\chi_{I_{0,0}}, \chi_{I_{1,0}}, \chi_{I_{1,1}}, \chi_{I_{2,0}}, \ldots$. If $f_n = \chi_{I_{k,j}}$ then $\|f_n\|_1 = 2^{-k}$, and $k \to \infty$ as $n \to \infty$, so $\|f_n\|_1 \to 0$. But for every $x \in [0, 1]$

and every level k, x lies in at least one $I_{k,j}$ and outside at least one other (for $k \geq 1$). So $f_n(x) = 1$ infinitely often and $f_n(x) = 0$ infinitely often; $f_n(x)$ does not converge.

(b) Choose $n_1 < n_2 < \cdots$ with $\mu(E_k) < 2^{-k}$, where $E_k = \{|f_{n_k} - f| \geq 2^{-k}\}$ (possible since $f_n \to f$ in measure). Since $\sum_k \mu(E_k) < \infty$, Borel–Cantelli says that a.e. x lies in only finitely many E_k. For such x there is K with $|f_{n_k}(x) - f(x)| < 2^{-k}$ for all $k \geq K$, so $f_{n_k}(x) \to f(x)$.

13.

For fixed $x \in (0, 1]$, note that

$$\frac{\partial}{\partial y} \frac{y}{x^2 + y^2} = \frac{x^2 + y^2 - 2y^2}{(x^2 + y^2)^2} = \frac{x^2 - y^2}{(x^2 + y^2)^2}.$$

So $\int_0^1 f(x, y)\, dy = \frac{1}{x^2+1}$, and $\int_0^1 \int_0^1 f\, dy\, dx = \int_0^1 \frac{dx}{1+x^2} = \frac{\pi}{4}$. Since $f(y, x) = -f(x, y)$, the other order gives $-\frac{\pi}{4}$.

No contradiction: Fubini requires $f \in L^1([0, 1]^2)$, and it is not. By Tonelli, using polar coordinates on the quarter-disc $\{r \leq 1,\ 0 \leq \theta \leq \pi/2\} \subseteq [0, 1]^2$, where $|f| = \frac{|\cos 2\theta|}{r^2}$,

$$\iint_{[0,1]^2} |f| \geq \int_0^{\pi/2} |\cos 2\theta|\, d\theta \int_0^1 \frac{r\, dr}{r^2} = 1 \cdot \infty = \infty.$$

14.

Write $\tau_h f(x) = f(x + h)$; note $\|\tau_h g\|_1 = \|g\|_1$ by translation invariance.

Continuous compactly supported case. Let $g \in C_c(\mathbb{R})$ with support in $[-R, R]$. Then g is uniformly continuous, so $\omega(h) = \sup_x |g(x + h) - g(x)| \to 0$ as $h \to 0$. For $|h| \leq 1$, $\tau_h g - g$ vanishes outside $[-R - 1, R + 1]$, so $\|\tau_h g - g\|_1 \leq (2R + 2)\, \omega(h) \to 0$.

General case. Given $\varepsilon > 0$ pick $g \in C_c$ with $\|f - g\|_1 < \varepsilon$. Then

$$\|\tau_h f - f\|_1 \leq \|\tau_h (f - g)\|_1 + \|\tau_h g - g\|_1 + \|g - f\|_1 < 2\varepsilon + \|\tau_h g - g\|_1,$$

so $\limsup_{h \to 0} \|\tau_h f - f\|_1 \leq 2\varepsilon$ for every ε.

Failure in L^∞. For $f = \chi_{[0,1]}$ and $0 < h < 1$, $\tau_h f - f = \chi_{[-h,0)} - \chi_{(1-h,1]}$ (up to endpoints), which has absolute value 1 on a set of positive measure. So $\|\tau_h f - f\|_\infty = 1$ for all small $h \neq 0$.

15.

(a) If $q = \infty$: $\int |f|^p \leq \|f\|_\infty^p \mu(X)$. If $q < \infty$: apply Hölder to $|f|^p \cdot 1$ with exponents $r = q/p > 1$ and $r' = q/(q - p)$:

$$\int |f|^p \leq \left(\int |f|^q \right)^{p/q} \mu(X)^{1 - p/q}.$$

Take p-th roots: $\|f\|_p \leq \mu(X)^{1/p - 1/q} \|f\|_q$.

(b) If $\mu(X) = 0$ or $\|f\|_\infty = 0$ both sides are 0. Otherwise: by (a) with $q = \infty$, $\|f\|_p \leq \mu(X)^{1/p} \|f\|_\infty$, and $\mu(X)^{1/p} \to 1$, so $\limsup_p \|f\|_p \leq \|f\|_\infty$. Conversely, for $0 < \varepsilon < \|f\|_\infty$ the set $A = \{|f| > \|f\|_\infty - \varepsilon\}$ has $\mu(A) > 0$ by definition of the essential supremum, and $\|f\|_p \geq (\|f\|_\infty - \varepsilon) \mu(A)^{1/p} \to \|f\|_\infty - \varepsilon$. So $\liminf_p \|f\|_p \geq \|f\|_\infty - \varepsilon$ for every ε.

16.

Lemma. A normed space is complete iff every absolutely convergent series converges. (If every such series converges and (x_n) is Cauchy, choose $n_1 < n_2 < \cdots$ with $\|x_{n_{k+1}} - x_{n_k}\| < 2^{-k}$; the series $x_{n_1} + \sum_k (x_{n_{k+1}} - x_{n_k})$ converges, so a subsequence of (x_n) converges, and a Cauchy sequence with a convergent subsequence converges.)

Let $f_k \in L^p$ with $\sum_k \|f_k\|_p = B < \infty$. Let $G_n = \sum_{k=1}^{n} |f_k|$ and $G = \sum_{k=1}^{\infty} |f_k| \in [0, \infty]$. By Minkowski, $\|G_n\|_p \le B$. Since $G_n^p \uparrow G^p$, MCT gives $\int G^p = \lim \int G_n^p \le B^p$. So $G \in L^p$ and in particular $G < \infty$ a.e. On the set where $G < \infty$ the series $\sum f_k(x)$ converges absolutely; define $F(x)$ as its sum there and 0 elsewhere. Then $|F| \le G$, so $F \in L^p$, and

$$\left| F - \sum_{k=1}^{n} f_k \right|^p \le (2G)^p \in L^1,$$

with the left side tending to 0 a.e. By DCT, $\|F - \sum_{k \le n} f_k\|_p \to 0$. By the lemma, L^p is complete.

17.

Let $P = \{f \ge 0\}$ and $N = \{f < 0\}$. For measurable $E \subseteq P$, $\nu(E) = \int_E f \ge 0$; for $E \subseteq N$, $\nu(E) \le 0$. So (P, N) is a Hahn decomposition.

The Jordan decomposition is $\nu^+(E) = \nu(E \cap P) = \int_E f^+ \, dm$ and $\nu^-(E) = -\nu(E \cap N) = \int_E f^- \, dm$, where $f^{\pm} = \max(\pm f, 0)$. These are mutually singular (concentrated on P and N respectively) and $\nu = \nu^+ - \nu^-$, so by uniqueness of the Jordan decomposition this is it. Then

$$|\nu|(E) = \nu^+(E) + \nu^-(E) = \int_E (f^+ + f^-) \, dm = \int_E |f| \, dm.$$

18.

If $c(E) = 0$ then $E = \emptyset$, so $m(E) = 0$. Hence $m \ll c$.

Suppose $m(E) = \int_E f \, dc$ for all Borel E. Taking $E = \{x\}$ gives $0 = m(\{x\}) = \int_{\{x\}} f \, dc = f(x)$ for every x. So $f \equiv 0$, and then $m([0,1]) = \int f \, dc = 0$, a contradiction.

The failing hypothesis: c is not σ-finite on $[0, 1]$ (a set of finite counting measure is finite, and a countable union of finite sets is countable).

19.

(a) Let $A = \limsup E_n = \bigcap_N \bigcup_{n \ge N} E_n$, the set of points in infinitely many E_n. For each N, $\mu(A) \le \mu(\bigcup_{n \ge N} E_n) \le \sum_{n \ge N} \mu(E_n)$, which tends to 0 as the tail of a convergent series. So $\mu(A) = 0$.

(b) Fix an integer k and work in $[k, k+1]$. For $q \ge 1$ let

$$E_q = \bigcup_{p \in \mathbb{Z}} \left(\frac{p}{q} - \frac{1}{q^3}, \frac{p}{q} + \frac{1}{q^3} \right) \cap [k, k+1].$$

Only p with $p/q \in [k-1, k+2]$ contribute, at most $3q + 1 \le 4q$ values of p. So $m(E_q) \le 4q \cdot \frac{2}{q^3} = \frac{8}{q^2}$, and $\sum_q m(E_q) < \infty$. By (a), a.e. $x \in [k, k+1]$ lies in only finitely many E_q. For

such an x, there are only finitely many q for which some p satisfies $|x - p/q| < q^{-3}$, and for each such q, $|qx - p| < q^{-2} \le 1$ allows at most two values of p (an open interval of length 2 contains at most two integers). So only finitely many pairs. A countable union over $k \in \mathbb{Z}$ of null sets is null.

Qualifying-exam level

20.

By Tonelli (for the counting measure on $\mathbb{Z}$ times Lebesgue measure on $[0,1]$; both σ-finite) and translation invariance,

$$\int_0^1 \sum_{n\in\mathbb{Z}} |f(x+n)|\,dx = \sum_{n\in\mathbb{Z}} \int_0^1 |f(x+n)|\,dx = \sum_{n\in\mathbb{Z}} \int_n^{n+1} |f| = \|f\|_1 < \infty.$$

So $\sum_n |f(x+n)| < \infty$ for a.e. $x \in [0,1]$. The function $x \mapsto \sum_n |f(x+n)|$ is 1-periodic, so the exceptional set $Z \subseteq \mathbb{R}$ satisfies $Z = Z + 1$, and $Z = \bigcup_{k\in\mathbb{Z}} ((Z \cap [0,1]) + k)$ is a countable union of null sets. So the series converges absolutely for a.e. $x \in \mathbb{R}$.

Since the double integral of $|f(x+n)|$ is finite, Fubini applies to $f(x+n)$:

$$\int_0^1 F(x)\,dx = \sum_{n\in\mathbb{Z}} \int_0^1 f(x+n)\,dx = \sum_{n\in\mathbb{Z}} \int_n^{n+1} f = \int_{\mathbb{R}} f.$$

21.

Let $\Lambda_n(f) = \int_0^{2\pi} f(x)|\sin nx|\,dx$ and $\Lambda(f) = \frac{2}{\pi}\int_0^{2\pi} f$. Both are linear and satisfy $|\Lambda_n(f)| \le \|f\|_1$, $|\Lambda(f)| \le \frac{2}{\pi}\|f\|_1 \le \|f\|_1$.

Step 1: intervals. Let $f = \chi_{[a,b]}$ with $0 \le a < b \le 2\pi$. The function $|\sin nx|$ has period π/n and $\int_0^{\pi/n} |\sin nx|\,dx = \frac{1}{n}\int_0^{\pi} \sin u\,du = \frac{2}{n}$, i.e. average value $\frac{2}{\pi}$ per period; by periodicity the integral over *any* interval of length π/n is $\frac{2}{n}$. Cut $[a,b]$, starting from a, into $N = \lfloor n(b-a)/\pi \rfloor$ consecutive intervals of length π/n, plus a leftover piece of length less than π/n on which the integrand is at most 1. So

$$\left| \int_a^b |\sin nx|\,dx - N \cdot \frac{2}{n} \right| \le \frac{\pi}{n},$$

and $N \cdot \frac{2}{n} = \frac{2}{\pi}(b-a) + O(1/n)$. Hence $\Lambda_n(\chi_{[a,b]}) \to \frac{2}{\pi}(b-a) = \Lambda(\chi_{[a,b]})$. By linearity, $\Lambda_n(s) \to \Lambda(s)$ for every step function s (finite linear combination of interval indicators).

Step 2: density. Step functions are dense in $L^1[0,2\pi]$. Given $\varepsilon > 0$ choose a step function s with $\|f - s\|_1 < \varepsilon$. Then

$$|\Lambda_n(f) - \Lambda(f)| \le |\Lambda_n(f-s)| + |\Lambda_n(s) - \Lambda(s)| + |\Lambda(s-f)| \le 2\varepsilon + |\Lambda_n(s) - \Lambda(s)|,$$

so $\limsup_n |\Lambda_n(f) - \Lambda(f)| \le 2\varepsilon$. As ε is arbitrary, the limit is $\Lambda(f)$.

(The same argument shows: if ϕ is bounded and T-periodic with mean c, then $\int f(x)\phi(nx)\,dx \to c\int f$ for every $f \in L^1$. The Riemann–Lebesgue lemma is the case $\phi(x) = e^{ix}$, $c = 0$.)

22.

For $a, b \in \mathbb{C}$ and $p \geq 1$, convexity of $t \mapsto t^p$ gives $|a-b|^p \leq (|a|+|b|)^p \leq 2^{p-1}(|a|^p + |b|^p)$. So

$$h_n = 2^{p-1}\left(|f_n|^p + |f|^p\right) - |f_n - f|^p \geq 0,$$

and $h_n \to 2^p|f|^p$ a.e. By Fatou,

$$\int 2^p|f|^p \leq \liminf_n \int h_n = 2^{p-1}\left(\lim_n \|f_n\|_p^p + \|f\|_p^p\right) - \limsup_n \int |f_n - f|^p = 2^p\|f\|_p^p - \limsup_n \|f_n - f\|_p^p.$$

Since $\|f\|_p < \infty$, subtract to get $\limsup_n \|f_n - f\|_p^p \leq 0$.

(The norm hypothesis is necessary: $f_n = n^{1/p}\chi_{(0,1/n)} \to 0$ a.e. with $\|f_n\|_p = 1 \not\to 0$.)

23.

$f \in L^2$: $|f_n|^2 \to |f|^2$ a.e., so by Fatou $\int |f|^2 \leq \liminf \int |f_n|^2 \leq 1$.

Weak convergence. Fix $g \in L^2[0,1]$ and $\varepsilon > 0$. Because $|g|^2 \in L^1$, the measure $E \mapsto \int_E |g|^2$ is absolutely continuous with respect to m and finite, so there is $\delta > 0$ with $m(E) < \delta \Rightarrow \int_E |g|^2 < \varepsilon^2$. (Proof of this standard fact: otherwise there are E_k with $m(E_k) < 2^{-k}$ and $\int_{E_k} |g|^2 \geq \varepsilon^2$; by Borel–Cantelli $A = \limsup E_k$ is null, yet by continuity from above for the finite measure $|g|^2 dm$, $\int_A |g|^2 = \lim_N \int_{\bigcup_{k \geq N} E_k} |g|^2 \geq \varepsilon^2$, a contradiction.)

All f_n and f are finite a.e. and $m([0,1]) < \infty$, so Egorov gives a measurable E with $m(E) < \delta$ and $f_n \to f$ uniformly on $[0,1] \setminus E$. Then, by Cauchy–Schwarz,

$$\left|\int_0^1 (f_n - f)g\right| \leq \sup_{[0,1]\setminus E} |f_n - f| \cdot \|g\|_1 + \|f_n - f\|_2 \left(\int_E |g|^2\right)^{1/2} \leq \sup_{[0,1]\setminus E} |f_n - f| \cdot \|g\|_2 + 2\varepsilon.$$

(We used $\|g\|_1 \leq \|g\|_2$ on $[0,1]$ and $\|f_n - f\|_2 \leq 2$.) The first term tends to 0, so $\limsup_n |\int (f_n - f)g| \leq 2\varepsilon$.

24.

Suppose $m(E) > 0$. Then $m(E \cap [-N, N]) > 0$ for some N; replace E by $E \cap [-N, N]$ (the hypothesis still holds for the smaller set). Apply the Lebesgue differentiation theorem to $\chi_E \in L^1(\mathbb{R})$: for a.e. x,

$$\lim_{r \to 0} \frac{m(E \cap (x-r, x+r))}{2r} = \chi_E(x).$$

Since $m(E) > 0$, there is a point $x \in E$ where this holds, so the ratio tends to 1. But by hypothesis the ratio is at most $c < 1$ for every r. Contradiction. Hence $m(E) = 0$.

25.

Recall: $C = \bigcap_k C_k$; $[0,1] \setminus C$ is a countable disjoint union of open intervals ("removed intervals"), and F is continuous, increasing, $F(0) = 0$, $F(1) = 1$, and constant on the closure of each removed interval.

(a) For a removed interval (a, b), $F(a) = F(b)$, so $\mu_F((a,b)) \leq \mu_F((a,b]) = F(b) - F(a) = 0$. By countable additivity, $\mu_F([0,1] \setminus C) = 0$ (note $\mu_F(\{0\}) = F(0) = 0$ is irrelevant since $0 \in C$). So μ_F is concentrated on C, while $m(C) = 0$ (Problem 4) and m is concentrated on $[0,1] \setminus C$. Hence $\mu_F \perp m$.

(b) F is constant on each removed interval, so $F' = 0$ on $[0,1] \setminus C$, i.e. a.e. If F were absolutely continuous, the fundamental theorem of calculus for Lebesgue integrals would give $1 = F(1) - F(0) = \int_0^1 F'\, dm = 0$. So F is not absolutely continuous. (Directly: for each k, C_k is a union of 2^k intervals of total length $(2/3)^k$ on which F increases by a total of 1.)

(c) The Cantor function satisfies $F(1-x) = 1 - F(x)$. (It holds at every stage of the standard construction $F = \lim F_k$, where F_k is piecewise linear, and C and its removed intervals are symmetric under $x \mapsto 1-x$; pass to the uniform limit.) Hence

$$\int_0^1 F(x)\, dx = \int_0^1 F(1-x)\, dx = \int_0^1 (1 - F(x))\, dx = 1 - \int_0^1 F(x)\, dx,$$

so $\int_0^1 F = \frac{1}{2}$.